EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Importance Of Chemistry, Nature Of Matter (Classification Of Elements, Compounds, Mixtures), Basic S.I Units

1. A pure substance can only be:

(A) A compound (B) An element

(C) An element or a compound (D) A heterogeneous mixture.

CBM077

Isotopes, Isobars, Isotones, Isodiaphers, Isostere, Isoelectronic (Calculation Of Electron, Proton and Neutron)

2. Find out the atoms which are isoneutronic :-

(A) ${}^{14}_{6}C, {}^{15}_{7}N, {}^{17}_{9}F$ (B) ${}^{12}_{6}C, {}^{14}_{7}N, {}^{19}_{9}F$ (C) ${}^{14}_{6}C, {}^{14}_{7}N, {}^{17}_{9}F$ (D) ${}^{14}_{6}C, {}^{14}_{7}N, {}^{19}_{9}F$

CBM079

3. Which of the following pairs is correctly matched :

(A) Isotopes, ${}^{40}_{20}Ca$, ${}^{40}_{19}K$ (B) Isotones, ${}^{30}_{14}Si$, ${}^{31}_{15}P$, ${}^{32}_{16}S$

(C) Isobars, ${}^{16}_{8}O$, ${}^{17}_{8}O$, ${}^{18}_{8}O$ (D) Isoelectronic, $N^{3\ominus}, O^{2\ominus}, Cr^{3\oplus}$

CBM081

Laws Of Chemical Combinations (Conservation Of Mass, Definite Proportion, Multiple Proportion, Reciprocal Proportion)

Rankers Academy JEE

4. In the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, ratio by volume of N_2, H_2 and NH_3 is 1 : 3 : 2. This illustrates law of

(A) Definite proportion (B) Multiple proportion

(C) Reciprocal proportion (D) Gaseous volumes

CBM085

5. The law of conservation of mass holds good for all of the following except -

(A) All chemical reactions (B) Nuclear reactions

(C) Endothermic reactions (D) Exothermic reactions

CBM087

6. If law of conservation of mass was to hold true, then 20.8 g of $BaCl_2$ on reaction with 9.8 g of H_2SO_4 will produce 7.3 g of HCl and $BaSO_4$ equal to :–

(A) 11.65 g (B) 23.3 g (C) 25.5 g (D) 30.6 g

CBM089

Dalton's Atomic Theory, Atomic Mass, Relative Atomic Mass(RAM), Gram Atomic Mass(GAM), Molecular Mass ,Relative Molecular Mass(RMM), Gram Molecular Mass (GMM)

7. What is the mass of a molecule of CH_4 :–

(A) 16 g (B) 26.6×10^{22} g (C) 2.66×10^{-23} g (D) $16\ N_A$ g

CBM092

8. Gram atomic weight of oxygen is

(A) 16 amu (B) 16 g (C) 32 amu (D) 32 g

CBM094

For More Join: @IITJEE_Advanced

Mole Concept (Basic Introduction To Mole, Mole Calculations)

9. The number of atoms present in 0.5 g atom of nitrogen is same as the atoms in –

(A) 12 g of C (B) 32 g of S (C) 8 g of oxygen (D) 24 g of Mg

CBM096

10. Number of oxygen atoms in 8 g of ozone is -

(A) 6.02×10^{23} (B) $\frac{6.02\times10^{23}}{2}$ (C) $\frac{6.02\times10^{23}}{3}$ (D) $\frac{6.02\times10^{23}}{6}$

CBM098

11. Which of the following contains the least number of molecules :-

(A) 4.4 g CO_2 (B) 3.4 g NH_3 (C) 1.6 g CH_4 (D) 3.2 g SO_2

CBM100

Ideal Gas Equation

12. Volume of 0.5 mole of a gas at 1 atm pressure and 273K is:

(A) 22.4 L (B) 11.2 L (C) 44.8 L (D) 5.6 L

CBM110

Empirical And Molecular Formulae

13. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. empirical formula of the compound is :-

(A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2

CBM106

14. A hydrocarbon contains 80% of carbon, then the hydrocarbon is -

(A) CH_4 (B) C_2H_4 (C) C_2H_6 (D) C_2H_2

CBM108

Pankees Academy JEE

Stoichiometry And Stoichiometric Calculations Without Limiting Reagent

15. The amount of water (g) produced by combustion of 286 g of propane is

(A) 168 g (B) 200 g (C) 468 g (D) 693 g

CBM115

16. Assuming that petrol is octane (C_8H_{18}) and has density 0.8 g mL^{-1}. 1.425 L of petrol on complete combustion will consume.

(A) 50 mole of O_2 (B) 100 mole of O_2 (C) 125 mole of O_2 (D) 200 mole of O_2

CBM117

17. The equation $2Al_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Al_2O_{3(s)}$ shows that :-

(A) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce $\frac{7}{2}$ mol of Al_2O_3

(B) 2 g of Al reacts with $\frac{3}{2}$ g of O_2 to produce one mol of Al_2O_3

(C) 2 g of Al reacts with $\frac{3}{2}$ L of O_2 to produce 1 mol of Al_2O_3

(D) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce 1 mol of Al_2O_3

CBM119

For More Join: @IITJEE_Advanced

Stoichiometry With Limiting Reagent And Excess Reagents

18. If 0.5 mol of $BaCl_2$ is mixed with 0.2 mol of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is -

$3BaCl_2 + 2Na_3PO_4 \longrightarrow Ba_3(PO_4)_2 + 6NaCl$

(A) 0.7 (B) 0.5 (C) 0.3 (D) 0.1

CBM170

19. 4 g of hydrogen is ignited with 4 g of oxygen. The weight of water formed is -

(A) 0.5 g (B) 3.5 g (C) 4.5 g (D) 2.5 g

CBM171

20. For the reaction A + 2B ⟶ C, 5 mol of A and 8 mol of B will produce

(A) 5 mole of C (B) 4 mole of C (C) 8 mole of C (D) 13 mole of C

CBM172

21. If 1.6 g of SO_2 and 1.5×10^{22} molecules of H_2S are mixed and allowed to remain in contact in a closed vessel until the reaction $2H_2S + SO_2 \longrightarrow 3S + 2H_2O$, proceeds to completion. Which of the following statement is true ?

(A) Only 'S' and 'H_2O' remain in the reaction vessel.

(B) 'H_2S' will remain in excess

(C) 'SO_2' will remain in excess

(D) None

CBM173

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Principle Of Atom Conservation (POAC)

22. 6 gm nitrogen on successive reaction with different compounds gets finally converted into 30 gm $[Cr(NH_3)_xBr_2]$ Value of x is [Atomic mass : Cr = 52, Br = 80]

(A) 1 (B) 2 (C) 3 (D) 4

CBM174

Percentage Yield , Percentage Purity

23. $CaCO_3$ is 90% pure. Volume of CO_2 collected at STP when 10 g of $CaCO_3$ is decomposed is -

(A) 2.043 L (B) 1.008 L (C) 10.08 L (D) 20.16 L

CBM175

24. 20 gm of a chalk sample is dissolved completely in excess of dilute HCl solution. If 3.36 L CO_2 gas is produced at 273K and 1atm, the percentage of pure $CaCO_3$ in chalk is (assume that the impurities are inert to HCl and CO_2 is completely insoluble in water)

(A) 25 % (B) 50 % (C) 75 % (D) 80 %

CBM176

Problems Based On Mixture, Sample, Impurity, And Heating Effect Using Mole Concept

25. Mixture of $MgCO_3$ & $NaHCO_3$ on strong heating gives CO_2 & H_2O in 3 : 1 mole ratio. The weight % of $NaHCO_3$ present in the mixture is:

(A) 30% (B) 80% (C) 40% (D) 50%

CBM177

For More Join: @IITJEE_Advanced

Parallel Reaction, Sequential Reaction

26. Carbon reacts with oxygen forming carbon monoxide and/or carbon dioxide depending an availability of oxygen. The respective moles of CO_2 obtained when 160 g oxygen reacts with (a) 12 g carbon (b) 120 g carbon (c) 72 g carbon will be

(A) 1, 0, 4 (B) 2, 0, 4 (C) 1, 0, 3 (D) 1, 1, 4

CBM179

27. 2 moles carbon and 1.5 moles of oxygen gas are reacted in a container to produce CO or CO_2 or both. The respective moles of CO, CO_2 produced will be

(A) 2, 1 (B) 1, 2 (C) 1, 1 (D) 2, 2

CBM180

Eudiometry (Gay Lussac's Law Of Combining Volumes)

28. 3 Litre mixture of C_2H_6 and C_3H_8 produce 8 litre CO_2 of complete combustion. Find the volume of C_2H_6 in mixture :-

(A) 2L (B) 1L (C) 1.5L (D) 2.5L

CBM181

29. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of $CO_2(g)$ and 50 mL of H_2O(vapour). The hydrocarbon is

(A) C_4H_5 (B) C_8H_{10} (C) C_4H_8 (D) C_4H_{10}

CBM182

Rankers Academy JEE

30. When 20 ml of mixture of O_2 and O_3 is heated, the volume becomes 29 ml and disappears in alkaline pyrogallol solution. What are the volume percent of O_2 in the original mixture?

(A) 90% (B) 10% (C) 18% (D) 2%

CBM183

31. 20 mL of a mixture of CO and H_2 were mixed with excess of O_2 and exploded & cooled. There was a volume contraction of 23 mL. All volume measurements correspond to room temperature (27°C) and one atmospheric pressure. Determine the volume ratio V_1: V_2 of CO and H_2 in the original mixture

(A) 6.5 : 13.5 (B) 5 : 15 (C) 9 : 11 (D) 7 : 13

CBM184

32. An alkene upon combustion produces $CO_2(g)$ and $H_2O(g)$. In this combustion process if there is no volume change occurs then the no. of C atoms per molecule of alkene will be :

(A) 2 (B) 5 (C) 3 (D) 4

CBM185

Concentration Terms (Percentage w/W, Percentage w/V, Percentage v/V,Grams Per Litre)

33. Unit of %w/w is :-

(A) $\frac{g}{litre}$ (B) $\frac{g}{ml}$ (C) $\frac{mol}{litre}$ (D) Unit less

CBM186

For More Join: @IITJEE_Advanced

34. 20 gm of $CaCO_3$ present in 80 gm of water, then find %w/w :

(A) 10% (B) 20% (C) 25% (D) 50%

CBM187

35. 200 gm solution of glucose by mass 25% is available in a vessel then find weight of glucose in a solution.

(A) 50 gm (B) 180 gm (C) 360 gm (D) 45 gm

CBM189

36. 20 gm $CaCO_3$ present in 80 ml of solution. Find $\% \frac{w}{V}$ of $CaCO_3$ solution :-

(A) 20% (B) 50% (C) 25% (D) None of these

CBM190

37. 500 ml solution having strength 20 $\frac{g}{litre}$. Find mole of solute in solution ($M_{w_{solute}} = 40$) :-

(A) 0.20 (B) 0.25 (C) 2.0 (D) 2.5

CBM192

38. Find % (w/v) of 20% (w/w) H_2SO_4 solution, if density of solution is 1.2 gram / ml.

(A) 24% (B) 12% (C) 20% (D) 16.6%

CBM195

39. 30 ml, 50% (v/v) HCl added into 70 ml, 20% (v/v) HCl solution. Find % (v/v) of resultant solution

(A) 14 (B) 29 (C) 15 (D) 70

Rankers Academy JEE

CBM196

40. 100 ml. aqueous solution (density = $\frac{5}{3}$ gm/ml) contain 40 % by weight NaOH. The number of molecules of NaOH dissolved in the above solution is :

(Use $N_A = 6 \times 10^{23}$)

(A) 2×10^{22} (B) 3.33×10^{22} (C) 10^{24} (D) 3.33×10^{23}

CBM197

Part Per Million(ppm), Part Per Billion (ppb)

41. 0.25 g of NaCl present in 999.75 g of water. Calculate ppm of NaCl solution :-

(A) 250 (B) 25 (C) 0.25 (D) 2.5

CBM198

Molarity, Mole Fraction, Mass Fraction, Molality

42. The Molarity of a solution of sodium chloride in water containing 5.85 g of sodium chloride in 500 ml of solution is:

(A) 0.25 M (B) 2.0 M (C) 1.0 M (D) 0.2 M

CBM199

For More Join: @IITJEE_Advanced

43. Molarity of liquid HCl, if density is 1.17 g/cc.:

(A) 36.5 M (B) 18.25 M (C) 32.05 M (D) 42.10 M

CBM200

44. Calculate molality of the solution obtained by dissolving 11.7 g NaCl in 500 g water

(A) 0.1 m (B) 0.3 m (C) 0.2m (D) 0.4m

CBM201

45. If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :

(A) 1 molal (B) 0.2 molal (C) 0.5 molal (D) 0.1 molal

CBM202

46. Molality of 720 g of pure water is :-

(A) 40 m (B) 4 m (C) 55.5 m (D) Can't be determine

CBM203

47. Mole fraction of glycerine ($C_3H_5(OH)_3$) in a solution of 36 g of water and 46 g of glycerine is :

(A) 0.46 (B) 0.36 (C) 0.20 (D) 0.40

CBM204

48. The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is :

(A) $\frac{8}{15}$ (B) 0.5 (C) 0.25 (D) 1.0

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CBM205

49. 18 gm of glucose present in 162 gm of H_2O. Calculate mass fraction of glucose :-

(A) 0.2 (B) 0.1 (C) 0.3 (D) 0.5

CBM206

50. Density of a solution containing 13% by mass of sulphuric acid is 0.98 g/ml. Then molarity of solution will be

(A) 1.1 M (B) 1.5 M (C) 1.3 M (D) 1.7 M

CBM207

51. The mole fraction of solute in aqueous urea solution is 0.2. The mass percent of solute is :

(A) 42.40 (B) 10.52 (C) 40.30 (D) 45.45

CBM208

52. The molarity of 98% by wt. H_2SO_4 (Density = 1.8 g/ml) is

(A) 6 M (B) 18 M (C) 10 M (D) 4 M

CBM209

53. Molality of 20% (w/w) aq. glucose solution is :

(A) $\frac{25}{18}$ m (B) $\frac{10}{9}$ m (C) $\frac{25}{9}$ m (D) $\frac{5}{18}$ m

CBM210

For More Join: @IITJEE_Advanced

54. Molarity and Molality of a solute (Molar weight = 50 g) in aqueous solution is 9 and 18 respectively. What is the density of solution.

(A) 1 g/cc (B) 0.95 g/cc (C) 1.05 g/cc (D) 2 g/cc

CBM211

55. 3.0 molal NaOH solution has a density of 1.12 g/ml. The molarity of the solution is-

(A) 2.97 M (B) 3 M (C) 3.05 M (D) 3.5 M

CBM212

Mixing And Dilution

56. 100 ml of 0.3 M HCl solution is mixed with 200 ml of 0.3 M H_2SO_4 solution. What is the molarity of $H^{\oplus}$ in resultant solution?

(A) 0.9 M (B) 0.6 M (C) 0.4 M (D) 0.5 M

CBM213

57. The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (18 M) with 50 ml of water, is:

(A) 36 M (B) 18 M (C) 9 M (D) 6M

CBM214

Volume Strength Of H_2O_2

58. Find the % w/v of "10 V" H_2O_2 solution-

(A) 2 % (B) 1 % (C) 3 % (D) 5 %

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CBM215

59. $2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

Under conditions where 1 mole of gas occupies 24 dm^3, X L $\frac{1}{24}$M of solution of H_2O_2 produces 3 dm^3 of O_2. Thus X is :-

(A) 8 (B) 10 (C) 15 (D) 6

CBM216

60. 34 g of hydrogen peroxide is present in 1135 ml of solution. Volume strength of solution is:

(A) 10 V (B) 20 V (C) 30 V (D) 32 V

CBM217

61. 20 ml of '20 vol' H_2O_2 solution is diluted to 80 ml. The final volume strength of solution is -

(A) 80 V (B) 25 V (C) 5 V (D) 8 V

CBM218

62. The strength of 11.35 volume solution of H_2O_2 is :

[Given that molar mass of H = 1 g/mol^{-1} and O = 16 g/mol^{-1}]

(A) 13.6% (B) 3.4% (C) 34% (D) 1.7%

CBM219

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Importance Of Chemistry, Nature Of Matter (Classification Of Elements, Compounds, Mixtures), Basic S.I Units

1. Which of the following is a mixture :

(A) Tap water (B) Distilled water

(C) Salt in water (D) Oil in water

CBM090

Laws Of Chemical Combinations (Conservation Of Mass, Definite Proportion, Multiple Proportion, Reciprocal Proportion)

2. Which of the following statement(s) is/are correct for water?

(A) H and O are in 2 : 1 atomic ratio.

(B) H and O are in 2 : 1 mass ratio.

(C) H and O are in 1 : 8 mass ratio.

(D) Hydrogen and oxygen gases are combined in 2:1 volume ratio.

CBM220

3. A quantity of 0.22 g of a gas occupies a volume of 112 ml at pressure of 1 atm and temperature of 273 K. The gas may be

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(A) nitrogen dioxide (B) nitrous oxide

(C) carbon dioxide (D) propane

CBM221

Limiting Reagent And Excess Reagents (Combustion Based Questions, Carbon Percentage, Hydrogen Percentage)

4. 0.2 mole of Cl_2 is reacted with 500 ml of 1.2 M - NaOH solution

$Cl_2 + OH^{\ominus} \rightarrow ClO_3^{\ominus} + Cl^{\ominus} + H_2O$

The incorrect information(s) related is/are -

(A) Cl_2 is limiting reagent

(B) 0.4 mole NaOH will left unreacted

(C) $\left[ClO_3^{\ominus}\right]_{\text{final solution}} = 0.133$ M

(D) $[Cl^{\ominus}]_{\text{final solution}} = 0.267$ M

CBM222

5. The oxygen needed for complete combustion of 8 g CH_4 may be obtained from complete decomposition of

(A) 2/3 mole of $KClO_3$ (B) 1 mole of H_2O_2

(C) 2 mole of $NaNO_3$ (up to 300oC) (D) 2 mole of BaO_2

CBM223

For More Join: @IITJEE_Advanced

6. When hydrocarbons are burnt completely in excess of oxygen gas, then

(A) equal moles of CO_2 and H_2O are formed from alkenes.

(B) more moles of H_2O than CO_2 are formed from alkanes.

(C) more moles of CO_2 than H_2O are formed from alkynes.

(D) more moles of CO_2 than H_2O are formed for any kind of hydrocarbon.

CBM160

Percentage Yield, Percentage Purity

7. Given

$2ZnS + 3O_2 \xrightarrow{80\%} 2ZnO + 2SO_2$

$ZnO + H_2SO_4 \xrightarrow{75\%} ZnSO_4 + H_2O$

$2ZnSO_4 + 2H_2O \xrightarrow{50\%} 2Zn + 2H_2SO_4 + O_2$

If 15 mol of ZnS reacts with excess of reactants then which of the following statement(s) is/are correct?

(A) 15 mol of Zn formed

(B) 2.25 mol of O_2 gas released

(C) 18 mol of H_2SO_4 formed

(D) 9 mol of $ZnSO_4$ formed

CBM224

Molarity, Mole Fraction, Mass Fraction , Molality

8. If the ratio of mole fractions of solute and solvent is unity, then the mass percent of solute is (Molar masses of solute and solvent are X and Y, respectively)

(A) 50%

(B) $\frac{X}{X+Y} \times 100\%$

(C) $\frac{X}{Y}$ × mass percent of solvent

(D) $\frac{Y}{X} \times$ mass percent of solvent

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CBM161

9. A quantity of 720 g water is added in 230g ethanol at a certain temperature to get 1 L of solution. Which of the following is/are correct regarding the solution formed?

(A) The density of solution is 950 kg/m^3.

(B) The mole fraction of ethanol is 0.11.

(C) The molarity of solution is 5 M.

(D) The molality of solution is 6.94 M.

CBM162

Mixing And Dilution

10. A dilute solution of H_2SO_4 is made by adding 5 mL of 3N H_2SO_4 to 245 mL of water. Find the normality and molarity of the diluted solution.

(A) Normality = 0.06 N

(B) Molarity = 0.03 M

(C) Normality = 0.03 N

(D) Molarity = 0.06 M

CBM225

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

Mole Concept (Basic Introduction To Mole, Mole Calculations)

11. Match the columns

Column-I		Column-II	
(A)	3 mole of $Co(NH_3)_4SO_4$	(P)	3 mole of S atom
(B)	1 mole of $FeKCo(NO_2)_6$	(Q)	1 mole of Fe atom
(C)	1.5 mole of $[Fe(H_2O)_5SCN]SO_3$	(R)	12 mole of O atoms
(D)	0.75 mole of $K_2Cu(SCN)_4$	(S)	6 mole of N atoms
		(T)	1.5 mole of K atoms

CBM163

12. Match the columns

Column-I (Average molecular mass)		Column-II (Composition of gas mixture)	
(A)	$\frac{80}{3}$	(P)	CH_4 and SO_3 gases in 1:1 mole ratio
(B)	$\frac{62}{3}$	(Q)	CH_4 and SO_3 gases in 1:1 mass ratio
(C)	48	(R)	CH_4 and SO_3 gases having H and O atoms in 8:3 ratio
(D)	$\frac{112}{3}$	(S)	CH_4 and C_2H_6 gases having C and H atoms in 2:7 ratio

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CBM164

Stoichiometry With Limiting Reagent And Excess Reagents (Combustion Based Questions, Carbon Percentage, Hydrogen Percentage)

13. Match the column-

Column-I		Column-II (Mass of product)	
(P)	$2H_2 + O_2 \rightarrow 2H_2O$ 1g 1g	(1)	1.028 g
(Q)	$3H_2 + N_2 \rightarrow 2NH_3$ 1g 1g	(2)	1.333 g
(R)	$H_2 + Cl_2 \rightarrow 2HCl$ 1g 1g	(3)	1.125 g
(S)	$2H_2 + C \rightarrow CH_4$ 1g 1g	(4)	1.214 g

(A) P-3; Q-4; R-1; S-2
(B) P-2; Q-4; R-1; S-3
(C) P-4; Q-3; R-1; S-2
(D) P-2; Q-3; R-1; S-4

CBM226

For More Join: @IITJEE_Advanced

Molarity, Mole Fraction, Mass Fraction, Molality

14. Match the column-

Column-I (Concentration of aqueous solution)		**Column-II (Density of given solutions is 1.2 g/ml)**	
(A)	2M NaOH solution	(P)	16 gm solute in 240gm solution
(B)	8% $\left(\frac{w}{V}\right)$ KOH solution	(Q)	60 gm solute in 240 gm solution
(C)	25% $\left(\frac{w}{W}\right)$ $CaCO_3$ solution	(R)	8 gm solute in 100 ml solution
(D)	$X_{C_3H_7OH} = \frac{1}{11}$	(S)	30 gm solute in 100 ml solution
		(T)	1 mole solute in 400 gm solution

(A) A-P,R; B-P,R; C-Q,S,T; D-S,Q
(B) A-P,S; B-R,S; C-Q,S,T; D-P,S
(C) A-P,T; B-S,T; C- S,T; D-S,Q
(D) A-R,S; B-P,S; C-Q,T; D-Q,R

CBM227

15. Match the columns-

Column-I		**Column-II**	
(A)	5m of NaOH solution ($d_{solution}$ = 0.6 g/ml) Molarity of solution is	(P)	16 M
(B)	250 ml of H_2O_2 solution provides 64g of O_2. Molarity of H_2O_2 solution is	(Q)	1 M
(C)	100 ml of 1 M-H_2SO_4 solution ($d_{solution}$ = 1.5g/ ml) is mixed with 400 ml of water, density of final solution = 1.25 g/ml. Molarity of resulting solution is	(R)	2.5 M
(D)	100 ml of 6 M-NaCl solution is mixed with 100 ml of 17% (w/w) $AgNO_3$ solution ($d_{solution}$ = 8 g/ ml). Molarity of Ag^+ ions in the resulting solution is	(S)	0.227 M

CBM165

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EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Dalton's Atomic Theory, Atomic Mass, Relative Atomic Mass(RAM), Gram Atomic Mass(GAM), Molecular Mass ,Relative Molecular Mass(RMM), Gram Molecular Mass (GMM)

1. Each molecules of a compound contains 5 carbon atoms, 8 hydrogen atoms and 4×10^{-23} gm of other elements. The molecular mass of compounds is ($N_A = 6 \times 10^{23}$)

CBM158

Mole Concept (Basic Introduction To Mole, Mole Calculations)

2. One atom of an element X weighs 6×10^{-23} gm. Number of moles of atoms in 3.6 kg of sample of X will be : ($N_A = 6 \times 10^{23}$)

CBM156

Principle Of Atom Conservation (POAC)

3. 3 moles of S_8 was treated by a series of reagents so as to convert all of its S to H_2SO_4. Calculate weight of product.

CBM228

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Percentage Yield, Percentage Purity

4. A 2000 gm sample of $CaCO_3$ is 80% pure. Find weight (in gm) of pure $CaCO_3$.

CBM229

5. An impure sample (having 60% purity) of $KClO_3$ contains 30 gm of pure $KClO_3$. Find weight of impure sample.

CBM230

Problems Based On Mixture, Sample, Impurity, And Heating Effect Using Mole Concept

6. 1.5 gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighing 1.46 gm. The reaction responsible for loss of weight is

$$Fe_2O_3(s) \longrightarrow Fe_3O_4(s) + O_2(g)$$

What is the percentage by mass of Fe_2O_3 in original sample.

CBM231

Parallel Reaction, Sequential Reaction

7. How many grams H_2SO_4 can be obtained from 1320 g PbS as per reaction sequence?

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

$$3SO_2 + 2HNO_3 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO$$

[At. mass : Pb = 208, S = 32]

CBM232

For More Join: @IITJEE_Advanced

8. $A \longrightarrow 2B + C$

$3B \longrightarrow 2D$

Find moles of D produced if initially 3 moles of A are taken –

CBM233

Eudiometry (Gay Lussac's Law Of Combining Volumes)

9. If V mL of a gaseous hydrocarbon, after explosion with excess of oxygen, showed a contraction of 2.5 V mL and a further contraction of 2 V mL with caustic potash, Find the number of H-atom in 1 molecule of hydrocarbon.

CBM234

10. What volume (in ml) of $O_2(g)$ is needed for complete combustion of 40 ml ethane gas (C_2H_6) ?

CBM235

Volume Strength Of H_2O_2

11. The volume strength of 100 ml H_2O_2 solution which produce 5.6 litre of oxygen gas at 1 bar & 0°C.

CBM236

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Rankers Academy JEE

Mole Concept (Basic Introduction To Mole, Mole Calculations)

12. The number of F^- ions in 4.2 gm AlF_3 is (Al = 27, F = 19)

CBM157

Ideal Gas Equation: Relative Density, Vapour Density (Dulong Petits Law)

13. The vapour density of two gases are in the ratio 1 : 3. Their molecular masses are in the ratio of :-

CBM113

14. A gaseous mixture contains 40% H_2 and 60% He, by volume. What is the average molecular mass of mixture?

CBM102

15. Element B exist in nature in two isotopic form B-10 (20%) and B -11 (80%). What is average atomic mass of B ?

CBM104

16. In the following reaction, if 10 g of H_2 reacts with N_2. What will be the volume (in Litre) of NH_3 at STP.

$N_2 + 3H_2 \rightarrow 2NH_3$

CBM123

For More Join: @IITJEE_Advanced

17. For complete combustion of 1.12 L of butane (C_4H_{10}), the produced volume (in Litre) of CO_2 (g) at STP will be.

CBM124

18. Calculate the total number of atoms in 5.6L of SO_2 gas at 0°C and 1 atm :–

CBM159

Stoichiometry And Stoichiometric Calculations Without Limiting Reagent

19. Nitric acid is manufactured by the Ostwald process, in which nitrogen dioxide reacts with water.
$3\ NO_2\ (g) + H_2O\ (l) \rightarrow 2\ HNO_3\ (aq) + NO\ (g)$
How many grams of nitrogen dioxide are required in this reaction to produce 25.2 gm HNO_3 ?

CBM237

20. What total volume, in litre at 627°C and 82.1 atm, could be formed by the decomposition of 16 gm of NH_4NO_3 ? Reaction : $2\ NH_4NO_3 \rightarrow 2N_2 + O_2 + 4H_2O(g)$. (R : 0.0821 atm L/mole-K)

CBM238

21. Calculate mass of phosphoric acid (in gm) required to obtain 53.4g pyrophosphoric acid.
$2H_3PO_4 \rightarrow H_4P_2O_7 + H_2O$

CBM239

Stoichiometry With Limiting Reagent And Excess Reagents (Combustion Based Questions, Carbon Percentage, Hydrogen Percentage)

22. Potassium superoxide, KO_2, is used in rebreathing gas masks to generate oxygen :
$KO_2(s) + H_2O(l) \longrightarrow KOH\ (s) + O_2(g)$
If a reaction vessel contains 0.158 mol KO_2 and 0.10 mol H_2O, how many moles of O_2 can be produced?
(Round of the answer to two decimal places)

Pankers Academy JEE

CBM240

23. Carbon reacts with chlorine to form CCl_4. 36 gm of carbon was mixed with 142 g of Cl_2. If ratio of mass of CCl_4 produced and the remaining mass of excess reactant is y : 1, then value of y is

CBM241

24. A chemist wants to prepare diborane (B_2H_6) by the reaction
$6\ LiH + 8BF_3 \longrightarrow 6Li\ BF_4 + B_2H_6$
If he starts with 2.0 moles each of LiH & BF_3. How many moles of B_2H_6 can be prepared.

CBM242

Principle Of Atom Conservation (POAC)

25. A 5.00 gm sample of a natural gas, consisting of methane (CH_4), and ethylene, (C_2H_4) was burned in excess oxygen, yielding 44/3 gm of CO_2 and some H_2O as products. What mole percent of the sample was ethylene?

CBM243

26. All carbon atom present in $KH_3(C_2O_4)2.2H_2O$ weighing 7.62 gm is converted to CO_2. How many gm of CO_2 were obtained. (K = 39)

CBM244

For More Join: @IITJEE_Advanced

Percentage Yield, Percentage Purity

27. 245 gm of $KClO_3$ on heating yielded 64 gm O_2. Find % yield of reaction. (K = 39, Cl = 35.5)

CBM245

Problems Based On Mixture, Sample, Impurity, And Heating Effect Using Mole Concept

28. 19 gm mixture of $Na_2CO_3(s)$ and $NaHCO_3(s)$ on heating gives 2.2 gm CO_2 gas.

Find % $NaHCO_3$ (by mass) in mixture.

CBM246

Parallel Reaction, Sequential Reaction

29. $2A + 3B \longrightarrow 4C + D$

(excess)

$3C \longrightarrow 2E$

If in the above reaction 6 moles of E are produced, find moles of 'A' initially taken.

CBM248

30. Sulphur trioxide may be prepared by the following two reactions:

$S_8 + 8O_2(g) \rightarrow 8SO_2(g)$

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

If amount of SO_3 will be produced from 1 mol of S_8 is w g, then value of w/100 is.

CBM249

Rankers Academy JEE

Eudiometry (Gay Lussac's Law Of Combining Volumes)

31. 10 ml of CO is mixed with 25 ml air having 20% O_2 by volume. What would be the final volume (in ml) if none of CO and O_2 is left after the reaction?

CBM251

32. Calculate the volume (in ml) of CO_2 evolved by the combustion of 50 ml of a mixture containing 40% C_2H_4 and 60% CH_4 (by volume)

CBM252

33. 10 moles of a mixture of CO (g) and $CH_4(g)$ was mixed with 22 moles of O_2 gas and subjected to sparking. Find the moles of gas absorbed when the residual gases are passed through alc. KOH.

CBM253

34. 60 ml of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 ml of N_2 was formed, calculate % volume of no gas in the mixture.

CBM254

35. When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume (in ml)?

CBM255

36. A 20 ml mixture of C_2H_4 and C_2H_2 undergoes sparking in gas eudiometer with just sufficient amount of O_2 and shows contraction of 37.5 ml. The volume (in ml) of C_2H_2 in the mixture is.

CBM256

For More Join: @IITJEE_Advanced

Concentration Terms, Percentage w/W, Percentage w/V, Percentage v/V,Grams Per Litre

37. 15 g of methyl alcohol is present in 100 mL of solution. If density of solution is 0.90 g mL^{-1}, calculate the mass percentage of methyl alcohol in solution

CBM257

Molarity, Mole Fraction, Mass Fraction, Molality

38. Calculate the molarity of the following solutions :

(a) 4g of caustic (NaOH) soda is dissolved in 200 mL of the solution.

(b) 5.3 g of anhydrous sodium (Na_2CO_3) carbonate is dissolved in 100 mL of solution.

(c) 0.365 g of pure HCl gas is dissolved in 50 mL of solution.

CBM258

39. A 7.0 M solution of KOH in water contains 28% by mass of KOH. What is density of solution in gm/ml ? (K = 39)

CBM259

Mixing And Dilution

40. (a) Calculate number of milimoles of 4.9 g H_2SO_4

(b) 50 mL, 0.2 M NaOH is mixed with 20 mL, 0.1 M NaOH. Calculate number of milimoles of Na^+ in final solution.

(c) 3.65 g HCl, 9.80 g H_2SO_4 and 4.9 g H_3PO_4 are mixed together. Calculate the total milimoles of H^+ in the final solution.

(d) 10 mL, 0.1 M NaOH and 10 mL, 0.2 M $Ca(OH)_2$ are added to 80 mL water. Calculate final milimoles & molarity of $[OH^-]$ in final solution.

Rankers Academy JEE

CBM262

41. Find out the volume (L) of 98% w/w H_2SO_4 (density = 1.8 gm/ ml), must be diluted to prepare 12.6 litres of 2.0 M sulphuric acid solution.

CBM263

42. 500 ml of 2 M NaCl solution was mixed with 200 ml of 1/4 M NaCl solution. Calculate the molarity of NaCl in final solution.

CBM266

43. A solution containing 200 ml 0.5 M KCl is mixed with 50 ml 19 % $\frac{w}{V}$ $MgCl_2$ and resulting solution is diluted 8 times. Molarity of chloride ion is final solution is :

CBM267

44. A mixture containing equimolar amounts of $Ca(OH)_2$ and $Al(OH)_3$ requires 0.5 L of 4.0 M HCl to react with it completely. Total moles of the mixture is :

CBM268

Volume Strength Of H_2O_2

45. 500 ml of a H_2O_2 solution on complete decomposition produces 2 moles of H_2O. Calculate the volume strength of H_2O_2 solution?

CBM270

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. 2.8×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its 'x' mg sample. The mass of CO_2 taken initially is
Given : $N_A = 6.02 \times 10^{23}$ mol^{-1} **[JEE (Main) 2025]**
(1) 196.2 mg (2) 98.3 mg (3) 150.4 mg (4) 48.2 mg
CBM325

2. 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A(M_A) and molar mass of B(M_B) in kg mol^{-1} are : **[JEE (Main) 2019]**
(1) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$ (2) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
(3) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$ (4) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$
CBM271

3. The most abundant elements by mass in the body of a healthy human adult are :
Oxygen (61.4%) ; Carbon (22.9%), Hydrogen (10.0%) ; and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms is
(1) 15 kg (2) 37.5 kg (3) 7.5 kg (4) 10 kg
[JEE (Main) 2017]
CBM034

4. The number of chlorine atoms in 20 mL of chlorine gas at STP is____10^{21}.
(Round off to the Nearest Integer).
[Assume chlorine is an ideal gas at STP
R = 0.083 L bar mol^{-1} K^{-1}, $N_A = 6.023 \times 10^{23}$]
Rankers Academy JEE
[JEE (Main) 2021]
CBM047

5. In Carius method for estimation of halogens, 180 mg of an organic compound produced 143.5 mg of AgCl. The percentage composition of chlorine in the compound is _____ %.
[Given : molar mass in g mol^{-1} of Ag : 108, Cl = 35.5] **[JEE (Main) 2025]**
CBM326

6. During "S" estimation, 160 mg of an organic compound gives 466 mg of barium sulphate. The percentage of Sulphur in the given compound is _____ %.
(Given molar mass in g mol^{-1} of Ba : 137, S : 32, O : 16) **[JEE (Main) 2025]**
CBM327

7. The elemental composition of a compound is 54.2%, C, 9.2% H and 36.6% O. If the molar mass of the compound is 132 g mol^{-1}, the molecular formula of the compound is :
[Given : The relative atomic mass of C : H : O = 12 : 1 : 16] **[JEE (Main) 2025]**
(1) $C_4H_9O_3$ (2) $C_6H_{12}O_6$ (3) $C_6H_{12}O_3$ (4) $C_4H_8O_2$
CBM328

8. Combustion of glucose ($C_6H_{12}O_6$) produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is: [Molar mass of glucose in g mol^{-1} = 180]
[JEE (Main) 2024]
(1) 480 (2) 960 (3) 800 (4) 32
CBM166

For More Join: @IITJEE_Advanced

9. If a rocket runs on a fuel ($C_{15}H_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are:
(Given: density of the fuel is 0.756 g/mL)
(1) 1188 g and 1296 g (2) 2376 g and 2592 g
(3) 2592g and 2376 g (4) 4349 g and 3142 g

[JEE (Main) 2022]
CBM050

10. Complete combustion of 3 g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is _____.
(Round off to the Nearest Integer).
[Use : $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0]

[JEE (Main) 2021]
CBM048

11. For the following reaction, the mass of water produced from 445 g of $C_{57}H_{110}O_6$ is :
$2C_{57}H_{110}O_6(s) + 163O_2(g) \longrightarrow 114CO_2(g) + 110\ H_2O(l)$
(1) 495 g (2) 490 g (3) 890 g (4) 445 g

[JEE (Main) 2019]
CBM039

12. 0.01 mole of an organic compound (X) containing 10% hydrogen, on complete combustion produced 0.9 g H_2O. Molar mass of (X) is _____ g mol^{-1}. [JEE (Main) 2025]
CBM329

13. When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is ___. (Nearest integer)
Given :
Molar mass of Al is 27.0 g mol^{-1}
Molar mass of O is 16.0 g mol^{-1} [JEE (Main) 2025]
CBM330

14. Xg of benzoic acid on reaction with aq. $NaHCO_3$ release CO_2 that occupied 11.2 L volume at STP. X is ____ g. [JEE (Main) 2025]
CBM331

15. 0.5 g of an organic compound (X) with 60% carbon will produce _____$\times 10^{-1}$ g of CO_2 on complete combustion.

[JEE (Main) 2023]
CBM125

16. An organic compound gives 0.220 g of CO_2 and 0.126 g of H_2O on complete combustion. If the % of carbon is 24 then the % hydrogen is _____ $\times 10^{-1}$. (Nearest integer) [JEE (Main) 2023]
CBM272

17. 120 of an organic compound that contains only carbon and hydrogen gives 330g of CO_2 and 270g of water on complete combustion. The percentage of carbon and hydrogen, respectively are.
(1) 25 and 75 (2) 40 and 60 (3) 60 and 40 (4) 75 and 25

[JEE (Main) 2022]
CBM049

18. Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O. The percentage of oxygen in the organic compound is:
(1) 51.63 (2) 63.53 (3) 53.33 (4) 50.33

[JEE (Main) 2021]
CBM045

For More Join: @IITJEE_Advanced

19. The ratio of mass percent of C and H of an organic compound ($C_XH_YO_Z$) is 6 : 1. If one molecule of the above compound ($C_XH_YO_Z$) contains half as much oxygen as required to burn one molecule of compound C_XH_Y completely to CO_2 and H_2O. The empirical formula of compound $C_XH_YO_Z$ is

(1) C_2H_4O (2) $C_3H_4O_2$ (3) $C_2H_4O_3$ (4) $C_3H_6O_3$

[JEE (Main) 2018]

CBM036

20. An unknown chloro-hydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only; chlorine atoms present in 1 g of chloro-hydrocarbon are :

(Atomic wt. of Cl = 35.5 u; Avogadro constant = 6.023×10^{23} mol^{-1})

(1) 6.023×10^{21} (2) 6.023×10^{23}

(3) 6.023×10^{20} (4) 6.023×10^{9}

[JEE (Main) 2018]

CBM038

21. The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is _____.

[JEE (Main) 2021]

CBM274

22. Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given : Density of nitric acid solution is 1.25 g/mL [JEE (Main) 2025]

(1) 45 (2) 55 (3) 32 (4) 40

CBM332

23. A solution is prepared by adding 2g of "X" of 1 mole of water. Mass percent of "X" in the solution is

(1) 20% (2) 5% (3) 2% (4) 10%

Rankers Academy JEE

[JEE (Main) 2023]

CBM126

24. A solution of H_2SO_4 is 31.4% H_2SO_4 by mass and has a density of 1.25g/mL. The molarity of the H_2SO_4 solution is M (nearest integer)

[Given molar mass of H_2SO_4 = 98g mol^{-1}]

[JEE (Main) 2024]

CBM167

25. Molality of 0.8 M H_2SO_4 solution (density 1.06 g cm^{-3}) is ____$\times 10^{-3}$ m.

[JEE (Main) 2024]

CBM168

26. The mass of sodium acetate (CH_3COONa) required to prepare 250 mL of 0.35 M aqueous solution is _____ g. (Molar mass of CH_3COONa is 82.02 g mol^{-1})

[JEE (Main) 2024]

CBM169

27. 250 g solution of D-glucose in water contains 10.8% of carbon by weight. The molality of the solution is nearest to

(Given: Atomic Weights are H, 1u ; C, 12u ; O, 16u) [JEE (Main) 2022]

(1) 1.03 (2) 2.06 (3) 3.09 (4) 5.40

CBM275

28. The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is _____ m.

(Round off to the Nearest Integer)

[Use : Atomic masses : Na : 23.0 u O : 16.0 u, H : 1.0 u

Density of H_2O : 1.0 g cm^{-3}] [JEE (Main) 2021]

CBM276

For More Join: @IITJEE_Advanced

29. An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1}. The molarity of the solution in mol L^{-1} is _____ (Nearest integer) **[JEE (Main) 2021]**
[Molar mass of KCl = 74.5]

CBM277

30. The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($H_2C_2O_4.2H_2O$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)
[Atomic mass : H : 1.0, C : 12.0, O : 16.0]

[JEE (Main) 2021]

CBM278

31. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times 10^{-1}$ M. (Nearest integer)
[Atomic mass : Na = 23.0, O = 16.0, H = 1.0]

[JEE (Main) 2021]

CBM279

32. If 80 g of copper sulphate $CuSO_4 \cdot 5H_2O$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3}$ mol L^{-1}. The value of x is _____.
[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u]

[JEE (Main) 2021]

CBM280

33. The strengths of 5.6 volume hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are:
(Take molar mass of hydrogen peroxide as 34 g/mol)

[JEE (Main) 2020]

(1) 1.7 and 0.25
(2) 1.7 and 0.5
(3) 0.85 and 0.5
(4) 0.85 and 0.25

CBM281

34. Density of 3 M NaCl solution is 1.25 g/mL. The molality of the solution is :

[JEE (Main) 2025]

(1) 1.79 m
(2) 2 m
(3) 3 m
(4) 2.79 m

CBM333

35. 20 mL of 2 M NaOH solution is added to 400 mL of 0.5 M NaOH solution. The final concentration of the solution is _____ $\times 10^{-2}$ M. (Nearest integer).

[JEE (Main) 2025]

CBM334

Bankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The stoichiometric reaction of 516 g of dimethyldichlorosilane with water results in a tetrameric cyclic product **X** in 75% yield. The weight (in g) of **X** obtained is___.
[Use, molar mass (g mol^{-1}): H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

[JEE (Advanced) 2023]

CBM282

2. The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $NiCl_2.6H_2O$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952g of $NiCl_2.6H_2O$ are used in the preparation, the combined weight (in grams) of gypsum and the nickel-ammonia coordination compound thus produced is___.
(Atomic weights in g mol^{-1}: H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59)

$(NH_4)_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4.2H_2O + 2NH_3$

$NiCl_2 \cdot 6H_2O + 6NH_3 \longrightarrow [Ni(NH_3)_6]Cl_2 + 6H_2O$

[JEE (Advanced) 2018]

CBM073

3. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of the molecular weights of the solute and solvent, $\left(\frac{MW_{solute}}{MW_{solvent}}\right)$, is

[JEE (Advanced) 2016]

CBM283

4. A compound **H_2X** with molar weight of 80 g is dissolved in a solvent having density of 0.4 g /ml, assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is.

[JEE (Advanced) 2014]

CBM284

5. Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives a gas **Q**. The amount of $CuSO_4$ (in g) required to completely consume the gas **Q** is _____.
[Given : Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63]

[JEE (Advanced) 2022]

CBM285

6. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is _____ .
(Atomic weights in g mol^{-1} : O = 16, S = 32, Pb = 207)

[JEE (Advanced) 2018]

CBM074

7. 29.2 % (w/w) HCl stock solution has a density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is.

[JEE (Advanced) 2012]

CBM286

8. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is

[JEE (Advanced) 2011]

(A) 1.78 M (B) 2.00 M (C) 2.05 M (D) 2.22 M

CBM287

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	D	B	B	C	B	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	C	C	C	C	D	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	D	A	C	D	A	C	B	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	A	D	B	A	C	B	A	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	D	C	D	D	C	C	B	B	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	B	A	B	B	D	C	C	D	A
Que.	61	62								
Ans.	C	B								

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	ACD	BCD	BD	ACD	ABC	BD	BC	ABCD	AB
Que.	11					12				13
Ans.	A → P, R; B → Q, R, S; C → P,R; D→P, T					A → Q; B → S; C → P; D → R				A
Que.	14	15								
Ans.	A	A → R; B → P; C → S; D → Q								

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EXERCISE (S)

1.	92	**2.**	100	**3.**	2352	**4.**	1600	**5.**	50
6.	80	**7.**	539	**8.**	4	**9.**	6	**10.**	140
11.	56.00	**12.**	9.03×10^{22}	**13.**	0.33	**14.**	3.20	**15.**	10.80
16.	75.66	**17.**	4.48	**18.**	$\frac{3}{4}N_A$	**19.**	27.60	**20.**	0.63
21.	58.80	**22.**	0.12	**23.**	6.41 or 6.42	**24.**	0.25	**25.**	33.33
26.	5.28	**27.**	66.67	**28.**	44.21	**29.**	4.5	**30.**	6.40
31.	30.00	**32.**	70.00	**33.**	10.00	**34.**	73.33	**35.**	10.00
36.	5.00	**37.**	16.66 or 16.67	**38.**	(a) 0.50 M, (b) 0.50 M, (c) 0.20 M			**39.**	1.40
40.	(a) 50 milimoles, (b) 12 milimoles, (c) 450 milimoles, (d) 5 milimoles, 0.05 M							**41.**	1.40
42.	1.5	**43.**	0.15	**44.**	0.80	**45.**	45.40		

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	3	1	20	40	3	2	3	18
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	100	153	61	11	56	4	3	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	8	3	4	4	815	7	2	5	3	20
Que.	31	32	33	34	35					
Ans.	13	64	2	4	57					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	
Ans.	222	2992	9	8.00	2.38/2.39	6.47	8	C	

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Important Notes

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